
Embeddings for Preferences, Not Semantics*

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Abstract

Modern AI is opening the door to collective decision-making in which participants express their views as free-form text rather than voting on a fixed set of candidates. A natural idea is to embed these opinions in a vector space so that the substantial literature on facility location problems and fair clustering can be brought to bear. But standard text embeddings measure semantic similarity, whereas distances in facility location problems and fair clustering require what we call *preferential similarity*: a participant’s agreement with a piece of text should be inversely related to their distance from it. Off-the-shelf embeddings inherit a coarse preference signal through a correlation between semantic and preferential similarity, but fail to capture preferences when the correlation breaks. We formalize this as an invariance problem: text embedding models encode both a preference-relevant signal (stance and values) and semantic nuisance (style and wording), and the two are observationally correlated, so a geometry that relies on nuisance can appear preference-correct even when it is not. We show that synthetic training data designed to break this correlation provably shifts the optimal scorer away from nuisance-dominated cosine and significantly improves preference prediction across 11 online deliberation datasets.

1 Introduction

Many emerging systems for collective decision-making allow participants to express their preferences in free-form text instead of voting on predetermined candidate sets. For example, two influential online deliberation platforms, Polis and Remesh, allow participants to write statements and vote on statements written by others [Small et al., 2021]. Similarly, in generative social choice, participants express their preferences using free-form text, which are then aggregated into a slate of representative statements [Fish et al., 2026]. A commonality among these systems is that they require methods to group participants and to estimate a participant’s utility for statements they did not vote on. Since inputs are free-form text, a natural idea is to embed each participant’s text using a text embedding model. This would permit the grouping of participants via clustering in the embedding space and distances could be used to estimate participants’ utility for statements they did not vote on. Further, it would allow for novel applications of ideas from the facility location and fair clustering literature [Feldman et al., 2016, Chen et al., 2019, Micha and Shah, 2020, Kellerhals and Peters, 2024].

However, off-the-shelf text embedding models are mainly trained and evaluated on semantic tasks such as retrieval, textual similarity, and natural-language inference [Muennighoff et al., 2023]. These tasks tend to reward placing texts close together in embedding space when they discuss the same topic or answer a similar query. They do not necessarily require close points to be mutually endorsable or, in other words, preferentially similar.

The distinction between semantic and preferential similarity is important to understand if embeddings are to be used inside preference aggregation procedures. To make the importance clear, it is useful to

*Code: <https://github.com/cartgr/Embeddings-for-Preferences>

Model: <https://huggingface.co/cartgr/embeddings-for-preferences-st5-xl>

Table 1: A hard triplet illustrating the mismatch between semantic and preferential similarity. The semantic distractor uses the anchor’s wording but reverses its stance, while the preference match preserves the stance but changes the wording. A base embedding model ranks the distractor closer to the anchor. A preference-tuned and topic-specific projected embedding produce the correct ranking.

Role	Text	Similarity		
		Base	Tuned (§5)	Projected (§7)
Anchor	“It should be legal . Religion has no place in politics if we truly want to be free.”			
Distractor	“It should be illegal . Religion has a place in politics if we truly want to be free.”	0.87	0.79	−0.74
Match	“To ensure true freedom, it’s crucial that legislation remains secular, without religious influence.”	0.82	0.81	0.99

imagine two statements from either side of a controversial political debate. They may share the same style and topic, as well as many of the same words, all of which a generic text embedding model may pick up on. However, they would not be mutually endorsed. Table 1 gives an example where a small alteration of a statement produces near-identical surface similarity to the anchor yet significantly alters the preferential similarity. A standard embedding model (ST5-XL) scores the altered distractor statement as more similar than an opinion-aligned alternative expressed in different words.

We frame the mismatch between semantic and preferential geometry as an invariance problem [Achille and Soatto, 2018]: a preference geometry should be invariant to wording and style, and sensitive only to stance and values. Generic embedding models do not have this invariance, because they encode topical and stylistic features that are useful for retrieval and similarity tasks but unrelated to whether two statements would be endorsed by the same person. On natural deliberation data, this gap is partly hidden because semantic and preferential similarity are correlated: people who share a stance often share wording. In §4.1 we make the confounding explicit by decomposing the cosine margin into a preference component and a nuisance component. Cosine weights the two equally, so when they agree (as on natural data) it looks correct, but when they disagree (as on the hard triplet in Table 1) the nuisance dominates and cosine fails.

Our method follows from this diagnosis. We synthesize triplets consisting of an anchor, a preference match with different wording, and a semantic distractor with high surface overlap but opposite stance. Training on these triplets forces cosine to down-weight nuisance variation that ordinary data leaves confounded with preference. We prove that, under this hard-triplet distribution, the Bradley-Terry risk is strictly decreased by reducing the nuisance contribution below cosine’s unit weighting. Empirically, our method significantly improves performance on hard triplets and on preference prediction across 11 online deliberation datasets.

When per-topic votes are available, which is common in online deliberation platforms, the same framework suggests an even simpler method: learning a low-rank projection of the frozen embedding. Despite its simplicity, this projected embedding performs very well in practice.

To summarize, our contributions are as follows:

1. We diagnose the mismatch between semantic and preferential similarity as an invariance problem, and formalize it via a decomposition of the cosine margin into preference signal and nuisance.
2. We introduce a hard-triplet synthesis procedure and prove that training on it strictly decreases the Bradley-Terry risk relative to standard cosine.
3. We demonstrate substantial gains on hard triplets and on preference prediction across 11 online deliberation datasets.
4. We show that when per-topic votes are available, a low-rank projection of frozen embeddings outperforms full preference tuning.

2 Related Work

A growing body of work on collective decision-making over free-form text relies on some geometry over participants or statements: Polis derives opinion maps from vote matrices [Small et al., 2021],

generative social choice groups statements in an LLM-defined feature space to produce representative slates [Fish et al., 2026], Blair et al. [2026] model consensus as a region of embedding space, and De et al. [2026] use cosine similarity of embeddings as participant utility when auditing justified representation in slates of questions. Our goal in this paper is to answer a prerequisite question. Namely, do distances in general-purpose sentence embedding spaces reflect preferential similarity? And, if not, can they be realigned so that they do?

Another line of work fine-tunes sentence encoders for opinion-related tasks, including stance-aware embeddings for opinion mining [Ghafouri et al., 2024] and sparsity-aware embeddings for contradiction retrieval [Xu et al., 2024]. However, in §6 we find that neither works well for our task. Our recipe is also related to SimCSE [Gao et al., 2021], which uses NLI entailments as positives and contradictions as hard negatives. Our task is different and we go a step further by engineering triplets where the nuisance signal intentionally points in the wrong direction (see Appendix E.4 for the relevant ablation). An extended related work discussion is in Appendix A.

3 Evaluation Setup

Before presenting our diagnosis and method, we describe the evaluation data and metrics used throughout. We evaluate on 11 datasets from three deliberation platforms. Generative social choice (GSC) [Fish et al., 2026] provides surveys in which participants write free-text opinions and then rate AI-generated statements (two abortion surveys and one on chatbot personalization). Remesh provides binary agree/disagree votes on others’ open-ended responses across three topics (campus protests, foreign intervention, right to assemble). Polis [Small et al., 2021] provides comment-level agree/disagree votes across five conversations (Seattle minimum wage, Bowling Green, Brexit, Canadian electoral reform, universal basic income). Some participants author comments in addition to voting. Together these span text lengths from short Polis comments to multi-paragraph GSC opinions and cover 1,462 participants, 3,958 statements, and 1.46M pairwise preference triplets. Dataset details and URLs are in Appendix B.

For each participant, we construct preference triplets (a, p, n) where a is the participant’s own written text (the anchor), p is a statement they rated more favorably, and n is one they rated less favorably. Candidate models score each triplet by computing a similarity margin $s(a, p) - s(a, n)$; the triplet is correct if this margin is positive. We call the fraction of correctly ordered triplets *triplet accuracy*, and *cosine accuracy* when the scorer s is cosine similarity. In §7 we also evaluate the ideal-point scorer, for which s is the learned distance-based utility.

4 Diagnosing Embedding Models

We begin by characterizing the preference signal in existing embedding models. We introduce a formal framework (§4.1), show that cosine is an approximation to the ideal-point utility margin, present empirical evidence for a correlation between nuisance and preference signal in the natural-data regime (§4.2), and diagnose the hard-triplet failure (§4.3).

4.1 Formal framework

Let $\psi : \mathcal{X} \rightarrow \mathbb{R}^d$ denote a pretrained encoder producing unit-norm embeddings. Within the embedding, preference on a given topic is governed by a *preference subspace* $S \subseteq \mathbb{R}^d$ of dimension $k \ll d$ that carries the stance-relevant structure. Let P_S denote orthogonal projection onto S and $P_{S^\perp}$ onto its complement, and abbreviate $\psi_S := P_S\psi$, $\psi_{S^\perp} := P_{S^\perp}\psi$. Each participant v writes anchor text a_v and has *ideal point* $u_v := \psi_S(a_v) \in S$, which is the projection of their own embedding onto S . Different participants have different anchors and thus different ideal points in S . For example, pro-choice and pro-life users live in opposite regions along the stance axis while sharing S as the relevant direction of variation.

We model the utility of candidate statement j as the negative squared Euclidean distance to the participant’s ideal point within S .² Expanding the square,

$$U^*(v, j) = -\|\psi_S(a_v) - \psi_S(x_j)\|^2 = 2\langle \psi_S(a_v), \psi_S(x_j) \rangle - \|\psi_S(a_v)\|^2 - \|\psi_S(x_j)\|^2.$$

²The derivations below extend to the general Mahalanobis case $-\|\psi_S(a_v) - \psi_S(x_j)\|_M^2$ for any PSD M on S : absorbing $\sqrt{M}$ into the encoder reduces it to the Euclidean case with no other change.

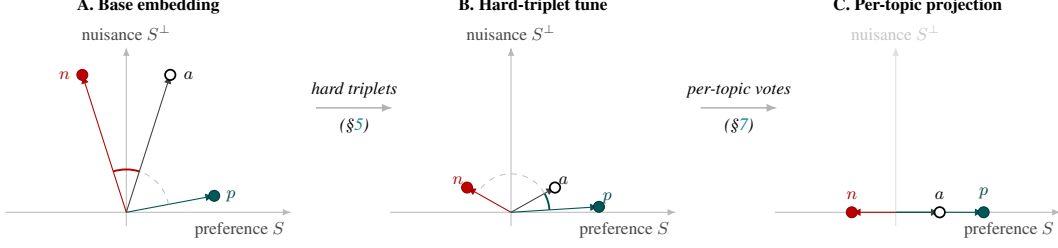


Figure 1: A hard triplet: anchor a , preference-match p (same stance, different wording), and semantic distractor n (opposite stance, same wording) with preference subspace S horizontal and nuisance $S^\perp$ vertical. (A) In the pretrained embedding, n shares a 's nuisance component, so cosine ranks n above p . (B) Fine-tuning on counterfactual hard triplets downweights $\psi_\perp$ (Theorem 1). (C) With per-topic labels, a rank- r projection $L^\top$ maps onto S directly, discarding $S^\perp$.

Pairwise preferences follow Bradley-Terry: $\Pr[p \succ n \mid v] = (1 + e^{-(U^*(v,p) - U^*(v,n))})^{-1}$. For within-anchor rankings, the $\|\psi_S(a_v)\|^2$ term is constant across candidates and cancels out, so the ranking-relevant utility margin is

$$U^*(v, p) - U^*(v, n) = \underbrace{2 \langle \psi_S(a_v), \psi_S(x_p) - \psi_S(x_n) \rangle}_{\Delta_S} + \underbrace{\|\psi_S(x_n)\|^2 - \|\psi_S(x_p)\|^2}_{\Delta_{\text{norm}}}. \quad (1)$$

The first term Δ_S is bilinear in the anchor and item projections and the second Δ_{norm} is an item-specific quadratic that depends only on candidate norms within S .

Cosine as an approximation. A cosine scorer on an off-the-shelf embedding model does not know S and uses all d dimensions equally. Because P_S and $P_{S^\perp}$ map into orthogonal subspaces, the cosine margin decomposes additively as

$$s(a_v, p) - s(a_v, n) = \underbrace{\langle \psi_S(a_v), \psi_S(x_p) - \psi_S(x_n) \rangle}_{\Delta_S} + \underbrace{\langle \psi_\perp(a_v), \psi_\perp(x_p) - \psi_\perp(x_n) \rangle}_{\Delta_T}. \quad (2)$$

Comparing (2) with (1), cosine captures the bilinear part of the utility margin (up to a factor of two) but misses the item-norm difference Δ_{norm} entirely, and contributes an out-of-subspace nuisance Δ_T that has no utility counterpart. Cosine is therefore an approximation to the ideal-point utility margin, tight when Δ_{norm} is small and when Δ_T aligns with Δ_S . In §7 we find empirical evidence that this model describes real preferences well.

Figure 1 illustrates the three stages in this paper: the diagnosis that base cosine overweights the nuisance direction (§4), hard-triplet fine-tuning which suppresses Δ_T (§5), and a per-topic rank- r projection which projects onto S directly (§7).

4.2 Proximity bands: cosine carries a preference signal

With the framework in place, we first ask what cosine similarity on base embedding models captures on natural deliberation data. This diagnostic is pairwise (between an anchor statement and a candidate statement) rather than triplet-based. For each participant, we compute cosine similarity from their anchor embedding to every statement they voted on, bin these pairs into quintiles by similarity, and compute the approval rate per band. We use the binary-vote datasets (Remesh and Polis), where approval is directly observed.

Approval rises monotonically across all four encoders (Figure 2). The spread between the most distant and most similar quintile is roughly between 15 and 20 percentage points for each encoder. From this it is clear that cosine carries some preference signal, as statements closer to a participant's own (according to cosine similarity) are more likely to be approved by that participant.

However, where this signal comes from is less clear. A high cosine score can arise because the candidate is close to the anchor in the preference subspace, or because it shares topic, wording, style, or affect that happen to correlate with preference in naturally occurring text on deliberation platforms.

More specifically, at the pair level similarity can be decomposed as

$$s(a_v, x_j) = \langle \psi(a_v), \psi(x_j) \rangle = \underbrace{\langle \psi_S(a_v), \psi_S(x_j) \rangle}_{s_S(a_v, x_j)} + \underbrace{\langle \psi_\perp(a_v), \psi_\perp(x_j) \rangle}_{s_T(a_v, x_j)}.$$

Figure 2 shows that the sum is predictive of approval but it does not identify whether the cause is the preference term s_S , the nuisance term s_T , or the two terms moving together. On natural deliberation data it could be the case that people who share a stance often also share wording and style. If this were the case, semantic similarity and preferential similarity would be observationally correlated.

The same ambiguity appears in the triplet margin used throughout the paper. For a triplet (a, p, n) with p preferred to n , differencing the pairwise scores gives

$$s(a, p) - s(a, n) = \Delta_S + \Delta_T.$$

Cosine gives Δ_S and Δ_T unit weight, but their effective strength on a dataset can be very different. The contribution of each term depends not only on the scorer’s coefficient, but also on the norm, variance, dimensionality, and label-alignment of the component in the observed triplets. If Δ_S is large and reliably aligned with approval, then cosine succeeds because the embedding already contains a strong preference geometry. If Δ_T is large and merely correlated with approval, then cosine can also succeed on natural data, but for the wrong reason — namely, by using semantic features that happen to track preferences in the observed distribution.

The proximity bands therefore leave open two explanations. Base embeddings may genuinely put nearby points near each other because they are preferentially similar. Alternatively, base embeddings may mostly capture semantic similarity, which may appear preference-aware only because semantic similarity is correlated with preferential similarity in data from online deliberation platforms. The next diagnostic is designed to ascertain which of the two explanations is correct by putting the two sources of signal in conflict.

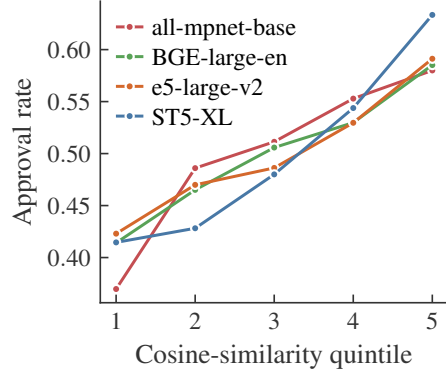


Figure 2: Approval rate by cosine-similarity quintile, pooled across the binary-vote evaluation datasets.

4.3 Hard triplets: separating preference from surface similarity

To test whether the apparent preference signal in base embedding models results from a correlated nuisance component or from a true preference awareness, we construct hard triplets that deliberately decouple preference alignment from surface similarity. For each anchor, we generate two comparison statements: a *semantic distractor* that shares the anchor’s vocabulary but takes the opposite stance, and a *preference match* that shares the anchor’s opinion but uses different vocabulary. A model that relies on surface features will score the distractor higher, whereas a model that captures preferential similarity will score the match higher.

We generate 875 such triplets (100 per evaluation dataset where available) by rewriting real participant anchors with GPT-4o (prompt in Appendix G). On these triplets, cosine fails across every encoder family we test. Base sentence-T5-XL reaches only 48.3%, near chance. Three widely-used encoders perform far worse: e5-large-v2 (26.7%), BGE-large (6.3%), and all-mpnet-base (8.2%). These three rank the semantic distractor above the preference match 70–95% of the time. Meanwhile these same encoders score 58–60% on the natural-data triplets (Table 2). The drops, of up to roughly 50 percentage points, from natural to hard data suggest that cosine’s apparent preference signal on natural data comes almost entirely from a nuisance component that is correlated with preferences on natural data.

In sum, hard triplets show that the natural-data signal from cosine on base embedding models is not robust when surface similarity and preference alignment conflict. This suggests using such conflicts not just as a diagnostic, but as supervision.

Table 2: Triplet accuracy (%) on the natural datasets (test split) vs. the hard triplets described in §4.3.

Encoder	Natural (mean, 11 ds.)	Hard ($n=875$)	Δ
sentence-T5-XL	65.2	48.3	-16.9
e5-large-v2	60.4	26.7	-33.7
BGE-large-en-v1.5	59.2	6.3	-52.9
all-mpnet-base-v2	57.8	8.2	-49.6

5 Decorrelated Preference Tuning

The hard-triplet diagnostic suggests a simple intervention: train on cases where surface similarity and preference alignment disagree. However, §4.2 and §4.3 suggest that these cases are rare in natural data. Therefore, we construct synthetic counterfactual triplets that break this correlation; we tune the encoder on them so that cosine learns to follow the preference-aligned item rather than the distractor that is similar in a superficial sense. We call this procedure *decorrelated preference tuning* (DPT).

5.1 Why hard triplets are corrective

We first formalize the direction in which hard-triplet training should move a scorer. Consider the class of bilinear scorers

$$s_A(a, x) = \psi(a)^\top A \psi(x),$$

with symmetric $A \in \mathbb{R}^{d \times d}$. To make the nuisance weight explicit, this can be written as

$$A(B, \lambda) = B + \lambda P_{S^\perp},$$

where $B = P_S B P_S$ only acts on the preference subspace and λ scales the nuisance. This family includes cosine on unit-norm embeddings when $B = P_S$ and $\lambda = 1$, and the S -projected scorer when $B = P_S$ and $\lambda = 0$.

For a triplet (a, p, n) , the margin is

$$\psi(a)^\top A(B, \lambda)(\psi(p) - \psi(n)) = \Delta_B + \lambda \Delta_T,$$

where $\Delta_B := \psi_S(a)^\top B(\psi_S(p) - \psi_S(n))$ and where $\Delta_T = \langle \psi_\perp(a), \psi_\perp(p) - \psi_\perp(n) \rangle$ is the nuisance margin. Thus, λ controls how much the scorer relies on the nuisance part of cosine.

Let $\mathcal{G} = \sigma(\psi_S(a), \psi_S(p), \psi_S(n))$ denote the σ -algebra generated by the in-subspace parts of the triplet. Conditioning on $\mathcal{G}$ means holding the preference subspace projections fixed while averaging over the nuisance subspace variation.

The hard-triplet condition can be formalized as $\mathbb{E}[\Delta_T \mid \mathcal{G}] \leq 0$ almost surely, with strict inequality on a set of positive probability. In words, once the preference-relevant parts of the triplet are fixed, the nuisance margin does not help the preferred item and, on average, it favors the distractor. Since we assume the preferred item in the hard triplet is aligned with the anchor in the preference subspace,³ this condition formalizes the idea that, in hard triplets, the preference and nuisance components point in opposite directions.

For the scorer $A(B, \lambda)$, define the Bradley-Terry population risk as

$$R(B, \lambda) = \mathbb{E}[\log(1 + e^{-\Delta_B - \lambda \Delta_T})].$$

The following result shows that, on any triplet distribution satisfying the hard-triplet condition, risk is reduced by decreasing the weight on the nuisance.

Theorem 1. *If $\mathbb{E}[\Delta_T \mid \mathcal{G}] \leq 0$ a.s., with strict inequality on a set of positive probability, then*

$$R(B, \lambda) < R(B, 1) \quad \text{for every } \lambda \in [0, 1).$$

The result does not require knowing S or directly optimizing λ . Instead, simply constructing a distribution of triplets where Δ_S points in the right direction and Δ_T points in the wrong direction for each triplet almost surely rewards scorers that put relatively more weight on the preference aligned part of the embedding. The proof is in Appendix C; its main steps are showing $R(B, \cdot)$ is convex, $R'(B, 0) > 0$, then using the convexity (and consequent non-decreasing derivative) to show that $R(B, \cdot)$ increases as λ moves from 0 toward 1.

³The following proof of Theorem 1, however, does not require this.

5.2 Counterfactual hard-triplet generation and fine-tuning

Theorem 1 suggests that training on a distribution of hard triplets should implicitly cause the encoder to place less weight on the nuisance subspace. We now describe how we synthesize such a distribution and how we train on it.

To begin, we assemble a pool of 2,000 political and social issues from the Habermas Machine dataset [Tessler et al., 2024] and Kialo,⁴ filtered by GPT-4o-mini for genuine policy debatability, and prompt Claude Sonnet 4 to generate 5 opinions per issue spanning the stance spectrum. We then sample anchors from this pool and prompt GPT-4o to rewrite each anchor into two versions. The *preference match* preserves the anchor’s stance and values but changes vocabulary, framing, and sentence structure. The *semantic distractor* preserves much of the anchor’s wording and structure but flips the stance. Together these form a triplet (a, p, n) in which the preferred item is stance-aligned but less surface-similar, while the dispreferred item is surface-similar but stance-opposed. Applying the same rewrite prompt to real participant anchors from the 11 evaluation datasets produces the held-out hard triplets used in §4.3 and §6. All prompts and model details are in Appendix G.

We remark that the training data is completely synthetic and was generated independently of the 11 benchmark datasets used for evaluation. So, any performance gains on the natural data benchmark are a strong indicator of generalization. In addition, the way the triplets are generated is specifically designed to approximate the hard-triplet condition, $\mathbb{E}[\Delta_T \mid \mathcal{G}] \leq 0$.

We adapt the pretrained encoder ψ with LoRA [Hu et al., 2022] ($r=16$, $\alpha=48$, targeting query and value projections). For a triplet (a, p, n) we train with Bradley-Terry loss over cosine differences:

$$\mathcal{L}_{\text{BT}}(a, p, n) = \log \left(1 + e^{-\left(\cos(\psi(a), \psi(p)) - \cos(\psi(a), \psi(n)) \right)} \right).$$

Training and inference both use cosine geometry, so what we optimize is what is used downstream. We do not train to convergence on the hard-triplet distribution since this would lead to negative weight on the nuisance subspace instead of the desired invariance.

6 Empirical Results

Fine-tuning substantially reduces the hard-triplet failure identified in §4.3. Our DPT-tuned sentence-T5-XL lifts hard-triplet (from the evaluation set) cosine accuracy from 48.3% to 80.0% and natural-triplet mean accuracy from 65.2% to 68.6%. Further, the recipe is not specific to sentence-T5: applying the same training to three widely-used BERT-derived encoders, e5-large-v2, BGE-large, and all-mpnet-base, produces similar patterns, with hard-triplet accuracy lifting by roughly 20 to 50 percentage points across models (Table 3). The natural-data lift is significant at the participant level under a paired Wilcoxon test ($p = 3.4 \times 10^{-9}$, pooled over the 442 test-split participants in the 11 datasets) and significant on 10 of 11 datasets under a triplet-level McNemar test (Appendix E.5).

Table 3: Base vs DPT-tuned cosine accuracy on the 11 natural-data datasets (test split) and on the 875 hard triplets across 5 training seeds. The recipe is hyperparameter-identical across models except for learning rate and hard-triplet count, which are val-selected per encoder.

Encoder	Natural (%)		Hard (%)		Gain (pp)	
	base	tuned	base	tuned	nat.	hard
sentence-T5-XL	65.2	68.6 ± 0.25	48.3	80.0 ± 0.37	+3.4	+31.7
e5-large-v2	60.4	63.0 ± 0.27	26.7	78.7 ± 0.54	+2.6	+51.9
BGE-large-en-v1.5	59.2	61.1 ± 0.15	6.3	32.8 ± 2.99	+1.9	+26.5
all-mpnet-base-v2	57.8	58.6 ± 0.16	8.2	27.4 ± 2.18	+0.8	+19.2

Table 4 shows where the resulting DPT-tuned geometry sits relative to a broader set of off-the-shelf and stance-aware baselines, including methods explicitly designed for viewpoint or contradiction-sensitive similarity. It shows a representative cut of the full 25-model comparison (Appendix F). Our DPT-tuned sentence-T5-XL achieves the highest mean accuracy at 68.6%, compared to 65.2% for the next-best embedding (the untuned ST5-XL). Our method is best on 8 of 11 datasets, and no other model is best on more than one.

⁴<https://huggingface.co/datasets/timchen0618/Kialo>

Table 4: Triplet accuracy (%) across the 11 evaluation datasets, abridged to seven representative models. A full comparison across 25 embedding models is in Appendix F.

Model	GSC			Remesh			Polis					Mean
	AbG	AbV	Chat	Camp	Frgn	RTA	Seatl	BG	Brxt	Can	UBI	
ST5-XL + DPT (ours)	81.8	76.4	65.9	67.5	65.6	69.5	69.8	65.1	68.4	62.5	61.9	68.6
ST5-XL	81.1	70.5	69.7	65.8	61.1	69.9	60.7	57.6	56.4	55.5	69.2	65.2
OAI text-emb-3-large	79.6	74.5	56.1	63.9	61.1	71.4	58.1	57.7	51.6	50.0	68.1	62.9
ST5-Large	75.5	65.5	57.2	64.6	59.9	69.2	58.8	58.1	49.9	56.8	72.2	62.5
BGE-large-en	64.7	59.5	56.5	62.5	57.6	68.0	54.2	55.5	53.2	52.9	66.3	59.2
StanceAware-SBERT	58.5	58.2	36.4	64.8	59.7	68.2	68.1	56.1	52.5	52.3	68.3	58.5
BGE-SparseCL (arguana)	62.4	61.7	42.7	59.2	57.2	66.1	50.1	56.1	57.5	52.6	63.0	57.2

7 Per-Topic Projected Embeddings

Decorrelated preference tuning trains a single encoder whose cosine geometry is more preference-aware without using any data from the target topic. This section asks what changes when participant votes over statements are available, as is often the case for online deliberation platforms. The section serves three purposes. First, it gives a stronger topic-specific deployment mode where the base embedding model is held fixed and a low-rank linear map is learned from participant votes. Second, it tests the ideal-point model from §4.1. And third, it sharpens the diagnosis of cosine similarity on base embedding models. If a low-rank linear map works on a frozen encoder, it suggests that the encoder already contains a preference signal. DPT makes this structure more visible to cosine by implicitly down-weighting the nuisance component, and the per-topic map recovers it from votes.

To begin, we fit the ideal-point utility directly on frozen sentence-T5-XL. Let a be a participant v 's anchor and x_j a candidate statement. We learn a rank- r linear map $L^\top : \mathbb{R}^d \rightarrow \mathbb{R}^r$ and score candidates by squared Euclidean distance in the mapped space:

$$U(v, j) = -\|L^\top \psi(a_v) - L^\top \psi(x_j)\|^2 = 2\langle L^\top \psi(a_v), L^\top \psi(x_j) \rangle - \|L^\top \psi(x_j)\|^2 - \|L^\top \psi(a_v)\|^2. \quad (3)$$

We refer to $\tilde{\psi}(x) = L^\top \psi(x)$ as a *projected embedding*. This form is the ideal-point model with the preference subspace and metric learned from votes. The same map is applied to anchors and candidates, and utility is distance rather than inner product. Fitting L separately for each of the 11 deliberation datasets with Bradley-Terry loss on participant votes (3-way train/validation/test split, rank $r = 20$, 5 seeds) reaches 77.6% macro-mean accuracy over the datasets on the test split. Applied to the hard evaluation triplets, with no training on hard data, the same per-topic linear map reaches 81.1% macro-mean accuracy (Table 5, first row). This suggests that the frozen encoder already contains preference signal and that L simply learns to read it out. This may also explain why DPT is so sample efficient — the geometry only needs to be realigned rather than learned from scratch.

We now turn to the question of whether the model proposed in §4.1 is empirically justified. The ideal-point scorer commits to three main structural claims about $\tilde{\psi}$: the scorer is *linear* in ψ , *tied* between anchor and item (both projected by the same L), and a *distance* rather than an inner product. Table 5 relaxes each claim in turn. Replacing the linear $L^\top$ with a shared nonlinear MLP ϕ while keeping the squared-distance form costs 4.1% on the natural data and 14.6% on the hard evaluation triplets, indicating that ψ is already rich enough that added per-text nonlinearity overfits rather than helps. Projecting anchor and item through independent L_a, L_j matches natural-data accuracy but loses 7.4% on the hard evaluation triplets. Dropping the item-quadratic $-\|L^\top \psi(j)\|^2$ for the bilinear cross-term alone (same rank, same shared L) leaves natural accuracy unchanged but loses 7.8% on

Table 5: Each row relaxes one structural commitment of the ideal-point scorer; all are fit on frozen base sentence-T5-XL with Bradley-Terry loss across the 11 evaluation datasets (3-way participant split, val-selected hyperparameters, five seeds).

Scorer on frozen ψ	Form	Nat. (%)	Hard (%)
Ideal-point scorer (ours)	$-\ L^\top \psi(a) - L^\top \psi(x)\ ^2$	77.6 ± 0.4	81.1 ± 0.5
Nonlinear [drops linearity]	$-\ \phi(\psi(a)) - \phi(\psi(x))\ ^2$	73.5 ± 0.6	66.5 ± 3.1
Asymmetric [drops tied projection]	$-\ L_a^\top \psi(a) - L_x^\top \psi(x)\ ^2$	77.0 ± 0.2	73.7 ± 0.4
Inner product [drops distance]	$\langle L^\top \psi(a), L^\top \psi(x) \rangle$	77.9 ± 0.3	73.3 ± 0.2

hard evaluation triplets. A possible explanation for this is that the item-norm encodes statement quality that directional similarity discards.

Our model also assumes that the learned space is low-dimensional. To test this we swept across ranks $r \in \{1, 2, 5, 10, 20, 50, 100\}$. Figure 3 shows that accuracy rises from $r = 1$ and plateaus at $r = 20$. Per-topic supervision is data-efficient as well. Fixing $r = 20$, the projected embedding crosses the universal DPT cosine at roughly 50 labeled triplets (Appendix E.7).

8 Discussion

This paper develops methods for learning a preference geometry over text, motivated by the design choices that deliberative systems must make. A key aspect of deliberation is exposing participants to a range of viewpoints [Mutz, 2006, Fishkin and Luskin, 2005, Sunstein, 2002]. Online platforms such as Polis, Remesh, Frankly, and the Stanford Online Deliberation Platform face this problem at scale, and must decide who should deliberate together, which comments to surface, and which viewpoints are missing [Small et al., 2021, Fishkin et al., 2019, Frankly, 2026]. A preference geometry gives one way to represent the viewpoints before making these decisions.

Two examples illustrate the point: forming deliberation groups, and aggregating the views expressed during deliberation. The composition of a deliberation group affects the quality of discussion, with quality highest at moderate levels of within-group disagreement [Esterling et al., 2015, Karpowitz and Mendelberg, 2007]. In practice, participants are assigned to small groups at random or by algorithms that balance demographics across groups alongside other objectives such as maximizing distinct pairwise meetings across rounds [Barrett et al., 2023, 2024], where demographic categories act as a coarse proxy for views [Mansbridge, 1999]. Yang and Bachmann [2025] move beyond demographics by using voting data as a proxy of views. A preference embedding space takes the final step, providing a direct measure of views in which objectives like balancing positions across rooms or controlling the level of within-room disagreement can be specified directly. Aggregation also has a natural formulation as voting in a metric space [Bulteau et al., 2021, Feldman et al., 2016]. Since platforms typically surface several comments rather than one, the relevant problem is selecting a slate, which corresponds geometrically to clustering. Here k -median and k -center capture utilitarian and egalitarian aggregation, while proportional fairness requires any large, aligned subgroup to have a representative [Chen et al., 2019, Micha and Shah, 2020, Kellerhals and Peters, 2024].

A preference geometry can also aid generative social choice [Fish et al., 2026, Boehmer et al., 2025], where generative models expand the set of candidate statements. Before generation, it can identify groups who deserve representation, during generation it can guide the generative model, and after generation, it can measure how well a generated statement represents its intended group. Appendix E.8 gives preliminary evidence that the tuned and projected geometries identify such groups better than the base embedding: on Remesh, users agree more with comments from their own cluster than from others, and the gap is significantly larger under the DPT-tuned and projected geometries.

Limitations and future work. First, our evaluation focuses on within-participant rankings, not whether absolute distances are calibrated, and since utility is latent, direct verification is difficult. Yet, Bradley-Terry training does push correctly-ordered pairs apart by a margin that scales with preference strength, and Appendix E.9 shows that similarity in the tuned and projected spaces tracks continuous Likert ratings better than the base model. Second, the gap between the universal tune and the per-topic projected embedding suggests that preferences have both a shared component across topics and a topic-specific component that only voting data on that topic recovers. A natural next step is to generate hard triplets conditioned on a target topic, producing a topic-specific embedding without per-topic votes. Finally, an embedding aimed at capturing preferences is an empirical and imperfect representation of preferences. It can support sense-making and help surface diverse views, but it should not be treated as a perfect representation of any individual’s considered judgment, and it should not be used as a basis for binding decisions [Revel and Pénigaud, 2026].

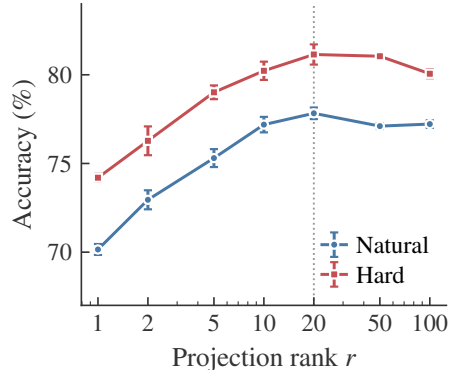


Figure 3: Per-topic scorer accuracy versus projection rank r . Mean $\pm$ std over five seeds, macro-averaged over 11 datasets.

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A Extended Related Work

Work on collective decision-making over free-form text increasingly relies on some representation of participants or statements. Polis derives a low-dimensional opinion map from the participant-by-comment vote matrix via PCA and clustering [Small et al., 2021, 2023], generative social choice and PROSE group statements in an LLM-defined feature space to produce representative slates [Fish et al., 2026, Boehmer et al., 2025], Blair et al. [2026] model approximate consensus as a region of sentence-embedding space, and De et al. [2026] audit justified representation of question slates using cosine similarity of question embeddings as participant utility. Complementary work aggregates text without an explicit embedding geometry, for example through reward models, token-level policies, or sampling-based social-choice procedures [Tessler et al., 2024, Blair and Larson, 2025, Chooi et al., 2026, Grandi et al., 2026].

A separate line of work studies the mismatch between semantic overlap and stance. Vahtola et al. [Vahtola et al., 2022] show that generic sentence embeddings struggle when negation or antonymy flips meaning; Introne [Introne, 2023] and Ghafouri et al. [Ghafouri et al., 2024] fine-tune encoders to separate opposing viewpoints for stance detection or opinion retrieval; and counter-argument retrieval methods add explicit dissimilarity terms or sparsity-aware scoring to retrieve contradictions [Wachsmuth et al., 2018, Xu et al., 2024]. We have a similar starting point, but make a unique diagnosis and propose a formal model, then use these to inspire a novel method.

A long tradition in political science estimates low-dimensional ideal points from political data. The canonical approach models legislators’ roll-call votes as a function of their positions in a latent spatial model [Poole and Rosenthal, 1985, Clinton et al., 2004]. A parallel line recovers positions directly from text: Wordscores scales manifestos against reference documents [Laver et al., 2003], Wordfish estimates time-series party positions from speech [Slapin and Proksch, 2008], and later work joins text with votes [Gerrish and Blei, 2011], combines topic models with ideal points [Vafa et al., 2020], or augments word embeddings with speaker metadata [Rheault and Cochrane, 2020]. Our §7 finding — that preferences on a frozen sentence embedding are linearly accessible through a rank-~20 projection — is consistent with this tradition, recovered here as a property of a general-purpose pretrained encoder on deliberation data from non-legislators.

Our work also connects to preference learning more broadly. Bradley-Terry objectives are standard in reward modeling and preference-based fine-tuning [Christiano et al., 2017, Ouyang et al., 2022, Rafailov et al., 2023], where they reshape a generative policy’s output distribution. Zhang et al. [2025] also embed responses but score preferences with a skew-symmetric operator to express within-user cyclic preferences, a concern that does not arise in our cross-user setting. We use the Bradley-Terry objective to reshape a reusable embedding geometry that can be consumed by metric social choice and other downstream geometric procedures [Anshelevich et al., 2021, Bulteau et al., 2021], rather than only a per-query reward model.

B Dataset Details

The GSC datasets come from two studies by Fish et al. [2026]. In each, participants first write free-text opinions describing their views on a topic, then rate a set of AI-generated policy statements on a Likert scale. We construct triplets from all pairs of statements with distinct ratings for each participant. The abortion generation survey uses participants who authored the original opinions; the abortion validation survey uses a separate cohort who rate the same statements and give verbal feedback to the statements. The chatbot personalization survey follows the same structure. Data for the abortion surveys is available at https://github.com/generative-social-choice/gsc_abortion and for chatbot personalization at https://github.com/generative-social-choice/chatbot_personalization.

The Remesh datasets come from the Polarized Issues corpus, which collects binary agree/disagree votes on open-ended responses to political questions using the Remesh platform. Participants first write their own response to a prompt, then vote on responses written by other participants. We use three topics: campus protests, foreign intervention, and right to assemble. The data is available at <https://github.com/akonya/polarized-issues-data>.

The Polis datasets come from publicly available Polis conversations from Small et al. [2021]. In Polis, participants write short comments and vote agree, disagree, or pass on comments written

Table 6: Evaluation datasets. Participants authored anchors (own text) and voted on statements (others’ text); pairwise preference triplets derive from their vote orderings. GSC uses a small fixed statement pool; Remesh and Polis use open pools seeded by participants themselves. We use all 11 datasets for evaluation.

Source	Dataset	Abbreviation	Participants	Anchors	Statements	Triplets
GSC	Abortion (generation users)	AbG	100	500	5	846
	Abortion (validation users)	AbV	100	993	10	3,513
	Chatbot Personalization	Chat	100	788	6	1,104
Remesh	Campus Protests	Camp	298	1,176	1,011	21,989
	Foreign Intervention	Frgn	289	857	764	12,428
	Right to Assemble	RTA	295	1,445	1,274	28,340
Polis	Seattle \$15/hour	Seatl	13	13	29	1,103
	Bowling Green AA	BG	222	531	607	1,363,191
	Brexit Consensus	Brxt	11	50	50	2,468
	Canadian Reform	Can	19	150	151	19,366
	NZ UBI	UBI	15	15	51	4,643
Total			1,462	6,458	3,958	1,458,991

by others. We restrict to participants who both authored at least one comment and voted on at least five others, using their authored comments as anchors. These datasets have the shortest texts and sparsest preference signal of any platform in our evaluation. The data is available at <https://github.com/compdemocracy/openData>.

Licenses. The GSC datasets are released under AGPL-3.0, the Polarized Issues (Remesh) data under CC-BY-4.0, and the Polis openData under CC-BY-4.0. The Habermas Machine dataset used for training-data generation is released under CC-BY-4.0 (data) and Apache-2.0 (code), and the Kialo dataset under MIT. The embedding models used are sentence-T5-XL (Apache-2.0), e5-large-v2 (MIT), BGE-large-en-v1.5 (MIT), and all-mpnet-base-v2 (Apache-2.0). All licenses permit research use.

C Proof of Theorem 1

Fix a symmetric in-subspace operator B with $B = P_S B P_S$, and write

$$\ell(u) = \log(1 + e^{-u}).$$

For a triplet (a, p, n) , recall that

$$\Delta_B = \psi_S(a)^\top B(\psi_S(p) - \psi_S(n)), \quad \Delta_T = \langle \psi_\perp(a), \psi_\perp(p) - \psi_\perp(n) \rangle.$$

Thus, the risk can be written as

$$R(B, \lambda) = \mathbb{E}[\ell(\Delta_B + \lambda \Delta_T)].$$

Our goal is to show that $R(B, \lambda) < R(B, 1)$ for every $\lambda \in [0, 1)$.

To do so, we will first note bounds which will be required below. Since the encoder outputs unit-norm embeddings and orthogonal projection cannot increase the norm (by the Pythagorean theorem),

$$\|\psi_\perp(a)\| \leq 1, \quad \|\psi_\perp(p) - \psi_\perp(n)\| \leq \|\psi_\perp(p)\| + \|\psi_\perp(n)\| \leq 2.$$

Therefore,

$$|\Delta_T| = |\langle \psi_\perp(a), \psi_\perp(p) - \psi_\perp(n) \rangle| \leq 2$$

everywhere. Also, since B is fixed and the embeddings have norm at most one,

$$|\Delta_B| \leq 2\|B\|_{\text{op}}.$$

Since $|\Delta_B + \lambda \Delta_T| \leq 2\|B\|_{\text{op}} + 2|\lambda|$ for each fixed λ , and ℓ is continuous, $\ell(\Delta_B + \lambda \Delta_T)$ is bounded and therefore integrable.

We now justify differentiating $R(B, \lambda)$ with respect to λ . The derivative of the logistic loss is

$$\ell'(u) = -\frac{1}{1 + e^u},$$

so $|\ell'(u)| \leq 1$ for every $u \in \mathbb{R}$. Fix $\lambda \in \mathbb{R}$ and, for nonzero h , define the difference quotient random variable

$$Q_h = \frac{\ell(\Delta_B + (\lambda + h)\Delta_T) - \ell(\Delta_B + \lambda\Delta_T)}{h}.$$

For each realization of the triplet, Δ_B and Δ_T are fixed finite numbers, and the map

$$\eta \mapsto \ell(\Delta_B + \eta\Delta_T)$$

is differentiable. Hence, as $h \rightarrow 0$,

$$Q_h \rightarrow \ell'(\Delta_B + \lambda\Delta_T)\Delta_T$$

pointwise. Moreover, by the mean value theorem, for each $h \neq 0$ there exists some ξ_h between λ and $\lambda + h$ such that

$$Q_h = \ell'(\Delta_B + \xi_h\Delta_T)\Delta_T.$$

Therefore

$$|Q_h| \leq |\ell'(\Delta_B + \xi_h\Delta_T)| |\Delta_T| \leq 2,$$

using $|\ell'| \leq 1$ and $|\Delta_T| \leq 2$. The constant function 2 is integrable, so dominated convergence gives

$$R'(B, \lambda) = \lim_{h \rightarrow 0} \frac{R(B, \lambda + h) - R(B, \lambda)}{h} = \lim_{h \rightarrow 0} \mathbb{E}[Q_h] = \mathbb{E}[\ell'(\Delta_B + \lambda\Delta_T)\Delta_T].$$

Thus, for every $\lambda \in \mathbb{R}$,

$$R'(B, \lambda) = \mathbb{E}[\ell'(\Delta_B + \lambda\Delta_T)\Delta_T].$$

We now evaluate this derivative at $\lambda = 0$. Let

$$\mathcal{G} = \sigma(\psi_S(a), \psi_S(p), \psi_S(n))$$

be the σ -algebra generated by the preference-subspace parts of the triplet. Since $B = P_S B P_S$, the quantity Δ_B is a function only of the projected variables in $\mathcal{G}$. Hence $\ell'(\Delta_B)$ is $\mathcal{G}$ -measurable. By the tower property,

$$R'(B, 0) = \mathbb{E}[\ell'(\Delta_B)\Delta_T] = \mathbb{E}[\mathbb{E}[\ell'(\Delta_B)\Delta_T \mid \mathcal{G}]].$$

Since $\ell'(\Delta_B)$ is $\mathcal{G}$ -measurable, the pull-out property gives

$$\mathbb{E}[\ell'(\Delta_B)\Delta_T \mid \mathcal{G}] = \ell'(\Delta_B) \mathbb{E}[\Delta_T \mid \mathcal{G}].$$

Therefore

$$R'(B, 0) = \mathbb{E}[\ell'(\Delta_B) \mathbb{E}[\Delta_T \mid \mathcal{G}]].$$

The hard-triplet condition states that

$$\mathbb{E}[\Delta_T \mid \mathcal{G}] \leq 0 \quad \text{a.s.,}$$

with strict inequality on a set of positive probability. Also,

$$\ell'(u) = -\frac{1}{1 + e^u} < 0 \quad \text{for every } u \in \mathbb{R},$$

and hence $\ell'(\Delta_B) < 0$ everywhere. Therefore the product

$$\ell'(\Delta_B) \mathbb{E}[\Delta_T \mid \mathcal{G}]$$

is nonnegative almost surely and strictly positive on a set of positive probability. Hence

$$R'(B, 0) > 0.$$

It remains to show that this positivity at zero implies the desired comparison with $\lambda = 1$. The loss ℓ is convex because

$$\ell''(u) = \frac{e^u}{(1 + e^u)^2} > 0 \quad \text{for every } u \in \mathbb{R}.$$

For each realization of the triplet, the map

$$\lambda \mapsto \Delta_B + \lambda \Delta_T$$

is affine, so

$$\lambda \mapsto \ell(\Delta_B + \lambda \Delta_T)$$

is convex. Taking expectations preserves convexity, and hence $R(B, \cdot)$ is convex. Since $R(B, \cdot)$ is differentiable, its derivative is nondecreasing. Therefore, for every $\lambda \in [0, 1]$,

$$R'(B, \lambda) \geq R'(B, 0) > 0.$$

Now fix any $\bar{\lambda} \in [0, 1)$. By the mean value theorem, there exists some $c \in (\bar{\lambda}, 1)$ such that

$$R(B, 1) - R(B, \bar{\lambda}) = R'(B, c)(1 - \bar{\lambda}).$$

Since $c \in [0, 1]$, we have $R'(B, c) > 0$, and since $1 - \bar{\lambda} > 0$,

$$R(B, 1) - R(B, \bar{\lambda}) > 0.$$

Equivalently,

$$R(B, \bar{\lambda}) < R(B, 1).$$

Because $\bar{\lambda} \in [0, 1)$ was arbitrary,

$$R(B, \lambda) < R(B, 1) \quad \text{for every } \lambda \in [0, 1).$$

□

D Additional Experimental Details

Hyperparameter selection. The sentence-T5-XL recipe reported in the main text was selected by a sweep over learning rate $\in \{1.25\text{e-}5, 5\text{e-}5, 1.25\text{e-}4, 5\text{e-}4\}$, hard-triplet count $\in \{250, 500, 750, 1000, 2000\}$, LoRA rank $\in \{4, 8, 16, 32, 64\}$, and LoRA $\alpha = 3r$ on a held-out validation split. The selected configuration is $\text{lr} = 1.25\text{e-}4$, $n_{\text{hard}} = 750$, $r = 16$, $\alpha = 48$, which corresponds to 46 gradient steps at batch size 16 on a random 750 of the 2,000-triplet synthetic pool, sampled uniformly without replacement and resampled independently for each training seed. For cross-model transfer (e5, BGE, all-mpnet) we fix rank and alpha and re-select $(\text{lr}, n_{\text{hard}})$ per encoder from the same grid; selected configurations are $(5\text{e-}5, 1000)$ for e5-large, $(1.25\text{e-}4, 500)$ for BGE-large, and $(1.25\text{e-}4, 1000)$ for all-mpnet. We AdamW with a linear-decay schedule and a 10% warmup ratio for all models.

Compute. All experiments were run on a single NVIDIA A100 GPU. Training a single DPT seed takes less than 20 minutes.

E Additional Experiments

E.1 Error Decomposition

To understand the residual errors of the tuned model, we classify a stratified sample of 913 errors using GPT-4o. Each error is a triplet where the tuned model incorrectly ranks the dispreferred statement higher. We present the full triplet (anchor texts, preferred, dispreferred) and ask the classifier to select from six categories, presented in randomized order to avoid position bias.

Surface similarity remains the dominant failure mode at 61%, which indicates room for further improvement through refined hard-triplet training. Insufficient anchor signal accounts for 20.5% and is concentrated in Polis datasets where comments are short: 73% of Bowling Green errors fall in this category, compared to 6% for GSC. Subtle value distinctions at 16.9% represent cases where both options broadly match the anchor’s views but differ on fine-grained nuance below the resolution of sentence-level similarity. The high number of “insufficient anchor signal” errors is notable since this is technically something no embedding model can solve. This must be addressed upstream by encouraging participants to write more or perhaps by giving them scaffolding to aid in their articulation of their preferences [Blair et al., 2025, Li et al., 2025, Handa et al., 2024].

Table 7: LLM-classified error categories for tuned model errors (913 triplets, stratified across datasets, capped at 5 per participant).

Category	Count	%
Surface similarity	557	61.0
Insufficient anchor signal	187	20.5
Subtle value distinction	154	16.9
Both options plausible	10	1.1
Style/register mismatch	3	0.3
None of the above	2	0.2

E.2 LoRA Rank Ablation.

We vary LoRA rank $r \in \{4, 8, 16, 32, 64\}$ with $\alpha = 3r$, holding all other hyperparameters at the selected ST5-XL configuration. Table 8 reports held-out test accuracy averaged across the 11 main evaluation datasets. Accuracy peaks at $r=16$: 67.6% at $r=4$, 68.7% at $r=16$, 67.1% at $r=32$, and 63.8% at $r=64$. With only 46 optimizer steps on 750 triplets, higher ranks introduce more free parameters than the data can constrain, and the extra capacity starts overwriting the pretrained geometry that the tuning is meant to realign, not replace.

Table 8: LoRA rank ablation on sentence-T5-XL (single seed, mean test accuracy across 11 datasets).

LoRA rank r	4	8	16	32	64
Mean test acc. (%)	67.6	68.3	68.7	67.1	63.8

E.3 Loss Ablation: Bradley-Terry vs InfoNCE.

We replace the pairwise Bradley-Terry loss with InfoNCE (temperature 0.05). For each anchor in a batch of 16, the positive is its own preference match and the negatives are its own semantic distractor plus the other 15 anchors’ preference matches as in-batch negatives, giving 16 negatives per anchor. All other hyperparameters are held at the selected ST5-XL configuration. The two losses yield essentially identical mean test accuracy (68.7% for BT vs 68.6% for InfoNCE), so the gain comes from the counterfactual triplet construction itself, not the choice of pairwise objective. We report Bradley-Terry in the main results because its score is exactly the signed cosine difference used at inference, and because it operates on each triplet independently — no batch size or batch composition to tune.

E.4 Normal-Correlation Triplet Ablation.

The hard-triplet construction in §4.3 engineers a worst-case correlation between stance and wording so that the nuisance margin satisfies the hard-triplet condition $\mathbb{E}[\Delta_T | \mathcal{G}] \leq 0$. The purpose of this is to force the model to rely less on the nuisance subspace and more on the preference subspace. We test whether this engineering is necessary by generating 2,000 *normal-correlation* training triplets in which stance and wording move together. This mimics more traditional NLI pair training where the correlation between the desired signal and a nuisance signal is not explicitly broken [Gao et al., 2021]. The rewrite prompt mirrors the hard prompt but flips the role assignment, so the preference match shares vocabulary with the anchor while the semantic distractor uses different vocabulary. The full prompt is in Appendix G. By construction $\mathbb{E}[\Delta_T | \mathcal{G}] > 0$, the opposite of the hard-triplet condition used in Theorem 1. We sweep the same hyperparameter grid used for the hard-triplet selection in Appendix D and select the configuration with the highest validation macro-mean accuracy on the 11 natural datasets, which gives $\text{lr} = 1.25\text{e-}5$, $n = 2000$, $r = 64$. Re-running this configuration over five seeds yields $67.0 \pm 0.1\%$ mean test accuracy and $59.3 \pm 0.2\%$ accuracy on the hard evaluation triplets. The matched hard-trained DPT configuration in Table 3 reaches $68.6 \pm 0.3\%$ test and $80.0 \pm 0.4\%$ on the hard evaluation triplets. Normal-correlation training therefore recovers a small lift over base cosine, but it is significantly worse than DPT.

E.5 Participant-Level Significance.

Table 3 reports dataset-mean accuracies across five training seeds, which is the right unit for comparing tuning recipes but does not directly speak to whether the gain is reliable on the unit of deployment, the individual participant. We therefore run a paired test on per-participant accuracies. For each of the 11 evaluation datasets we replay the test-split scoring loop under both base sentence-T5-XL and the DPT-tuned encoder, keeping the per-triplet outcome (1 if the encoder ranks the preference match above the distractor, 0 otherwise) and the per-participant accuracy (e.g., the accuracy for participant i using the base model is a_i^{base}). Each participant then has a pair $(a_i^{\text{base}}, a_i^{\text{tuned}})$, and each triplet has a pair of binary outcomes. We report a paired Wilcoxon signed-rank test on the participant accuracies and a McNemar exact test on the triplet outcomes.

Pooled across all 442 participants in the 11 test cohorts, the mean accuracy lift is +2.0 pp (median +0.9 pp). Tuned beats base on 51.6% of participants, ties on 17.2%, and loses on 31.2%. The paired Wilcoxon test yields $p = 3.4 \times 10^{-9}$ and the paired t-test yields $p = 1.8 \times 10^{-8}$. McNemar over the 24,834 discordant triplets returns $b = 14,891$ wins for tuned and $c = 9,943$ wins for base, with $p = 9.5 \times 10^{-218}$. Per-dataset results are in Table 9. The participant-level Wilcoxon reaches $p < 0.05$ on 5 of 11 datasets, while the triplet-level McNemar reaches $p < 0.05$ on 10 of 11. The lone non-mover under both is GSC chatbot, where DPT shows the smallest accuracy lift across the entire benchmark ($\Delta = +0.4$ pp). The Wilcoxon non-significance on the four small Polis cohorts (Seattle, Brexit, Canadian, UBI) reflects power rather than effect, since each shows tuned-mean above base-mean and the triplet-level McNemar reaches $p < 0.05$ on all four. The per-dataset and pooled accuracies in Table 9 are means of per-participant accuracies, which weight each participant equally rather than each triplet (as Table 15 and Table 4 do).

Table 9: Paired test of DPT-tuned vs base sentence-T5-XL at the participant and triplet level (test split, seed-42 DPT). Δ is the per-participant accuracy difference in percentage points. Wilcoxon is the paired signed-rank test on participant accuracies, McNemar is the exact binomial test on per-triplet wins and losses.

Dataset	n	base (%)	tuned (%)	Δ (pp)	Wilcoxon p	McNemar p
GSC Abortion (gen)	30	78.1	84.5	+6.4	4.3×10^{-2}	1.5×10^{-3}
GSC Abortion (val)	30	71.8	78.0	+6.2	9.9×10^{-3}	3.5×10^{-8}
GSC Chatbot (gen)	30	65.1	65.5	+0.4	7.5×10^{-1}	1.0
Remesh Campus Protests	90	63.7	65.9	+2.2	3.5×10^{-5}	1.3×10^{-12}
Remesh Foreign Interv.	87	62.2	63.3	+1.1	4.3×10^{-2}	3.4×10^{-3}
Remesh Right to Assemble	89	69.3	70.3	+1.0	7.5×10^{-2}	1.4×10^{-2}
Polis Seattle	4	70.4	72.9	+2.6	7.5×10^{-1}	4.1×10^{-2}
Polis Bowling Green	67	61.1	61.8	+0.7	3.0×10^{-2}	7.2×10^{-189}
Polis Brexit	4	69.9	72.0	+2.1	0.62	5.4×10^{-3}
Polis Canadian	6	50.7	57.1	+6.3	6.2×10^{-2}	6.9×10^{-10}
Polis UBI	5	55.0	58.0	+3.0	0.12	1.4×10^{-2}
Pooled	442	65.6	67.6	+2.0	3.4×10^{-9}	9.5×10^{-218}

E.6 Per-Topic Scorer: Rank Sweep

Figure 3 in §7 shows accuracy as a function of projection rank r . Table 10 reports the exact macro-mean values with standard deviations across five seeds.

Table 10: Per-topic scorer accuracy as a function of rank (macro-mean over 11 datasets, mean $\pm$ std over five seeds).

Rank r	1	2	5	10	20	50	100
Natural (%)	70.1 $\pm$ 0.3	72.9 $\pm$ 0.6	75.4 $\pm$ 0.5	77.1 $\pm$ 0.4	77.6 $\pm$ 0.4	77.1 $\pm$ 0.2	77.3 $\pm$ 0.2
Hard (%)	74.0 $\pm$ 0.4	76.4 $\pm$ 0.8	79.2 $\pm$ 0.4	80.4 $\pm$ 0.6	81.1 $\pm$ 0.5	81.0 $\pm$ 0.5	80.3 $\pm$ 0.4

E.7 Per-Topic Scorer: Data Efficiency

Fixing $r=20$, we vary the number of labeled triplets per topic and evaluate held-out natural accuracy. Figure 4 plots the learning curve across individual datasets and the macro-mean against the base and universal DPT-tuned cosine baselines. Averaging over datasets, $K \approx 50$ labeled triplets already cross the universal DPT baseline, and accuracy saturates by $K \approx 1,000$.

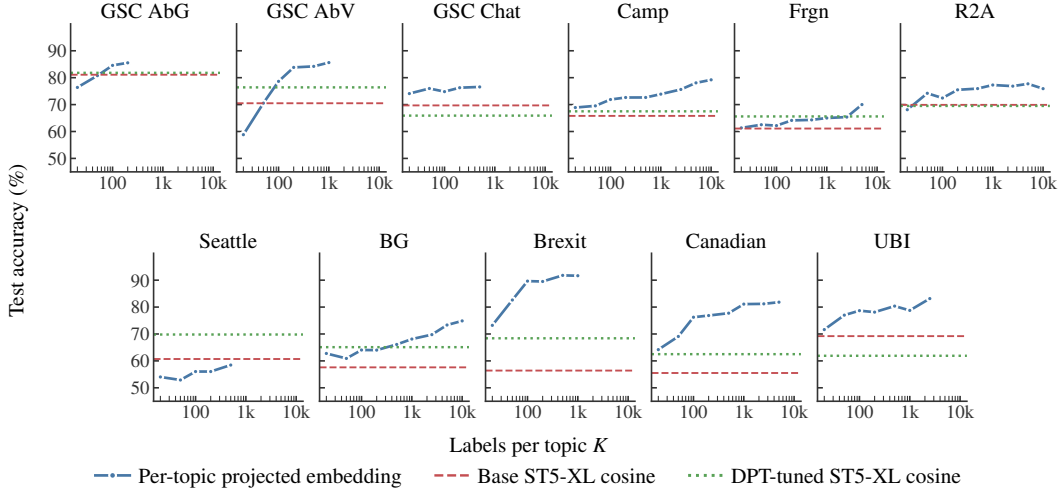


Figure 4: Data efficiency of the rank-20 per-topic projected embedding on base sentence-T5-XL.

E.8 User Clustering Coherence on Remesh

Here we test whether the DPT-tuned and per-topic projected embeddings provide a geometry in which clusters are coherent in approval behavior, beyond what the base ST5-XL geometry already provides. The Remesh deliberation transcripts are well suited to this question as each participant authors one or more comments, votes *Agree* or *Disagree* on a sample of comments written by others, and the comments tend to be more substantive than those from Polis. Further, in contrast to GSC, there are many statements that were voted on so the approval level for any cluster can be estimated. Taken together, this allows us to get another angle on how well the geometry aligns with participants’ expressed approval preferences in a setting akin to how it may be deployed.

For each participant u we form a user vector by mean-pooling the encoder embeddings of their authored comments. We do this under three geometries, the base sentence-T5-XL encoder, the DPT-tuned encoder of §5, and the rank-20 per-topic projection $L^\top \psi$ of §7, with L taken from the ideal-point scorer fit on the same dataset. We then run k -means on the user vectors for $k \in \{3, 5, 8, 10\}$. For every user u assigned to cluster c , we compute their *within-cluster approval rate*, defined as the fraction of *Agree* votes among the cluster- c -authored comments that u actually voted on. The denominator counts only votes that exist, and comments that were never shown to u are excluded. The analogous *across-cluster* rate uses the comments not in u ’s cluster.

We aggregate per-user rates within a cluster by vote-weighted mean and then macro-average over the clusters at a given k . The headline statistic is the lift $\Delta = \overline{\text{within}} - \overline{\text{across}}$, which is approximately zero under random assignment. Each cell in Table 11, indexed by dataset, encoder, and k , averages over five k -means seeds. We additionally compute a permutation null by shuffling cluster labels, with 50 permutations per seed. All shuffle lifts have absolute value at most 0.005, so we report raw lifts directly.

Table 11 reports the per-configuration lifts. Both adapted geometries exceed the base on a strong majority of the 12 cells. Clustering separates users most cleanly on Campus Protests, where $\Delta \approx 0.10$ for both the tuned encoder and the projection at all values of k . It separates them less cleanly on Foreign Intervention, where the projection collapses at $k \geq 8$, and only weakly on Right to Assemble, where the global within-rate is approximately 0.78 and there is little disagreement left to partition.

Table 11: Within–across approval-rate lift Δ (%) on Remesh user clusters, by encoder (base ST5-XL; DPT-tuned ST5-XL; rank-20 per-topic projection of base ST5-XL via $L^\top \psi$). Each cell is the mean of five k -means seeds; per-seed shuffle null has $|\Delta| \leq 0.5\%$. Best entry per row in **bold**.

Dataset	k	Base	Tuned	Proj
Campus Protests	3	9.7	11.1	9.5
	5	6.4	8.5	11.5
	8	8.7	11.0	11.1
	10	8.8	10.4	10.7
Foreign Intervention	3	2.4	5.8	10.1
	5	6.4	5.1	11.4
	8	7.7	9.0	0.7
	10	6.0	7.1	2.6
Right to Assemble	3	1.4	1.8	2.0
	5	2.8	3.7	2.9
	8	4.5	3.3	2.3
	10	2.1	3.4	3.9

Aggregating across the 12 configurations, the mean lift is 5.6% for base, 6.7% for DPT-tuned, and 6.6% for the per-topic projected embedding. The mean rank, with lower being better, is 2.50, 1.75, and 1.75 respectively. Paired comparisons over configurations give $\Delta_{\text{DPT-tuned}} - \Delta_{\text{base}} = +1.11$ pp, with 95% bootstrap CI $[+0.4, +1.8]$ and Wilcoxon $p = 0.016$, and $\Delta_{\text{projected}} - \Delta_{\text{base}} = +1.00$ pp, with CI $[-1.3, +3.1]$ and $p = 0.38$. The wider interval for the projection is driven by the two values of k on Foreign Intervention where the rank-20 projection collapses. DPT-tuned and the projected embedding are statistically indistinguishable at this scale, with $\Delta = +0.1$ pp, CI $[-1.9, +2.3]$, and $p = 0.97$. These results suggest that the DPT-tuned embedding provides a geometry which would be useful for the types of tasks identified in §8.

E.9 Likert-Rating Correlation on GSC

The triplet metric used throughout the paper collapses each participant’s continuous Likert rating of a candidate statement into a binary preference between two statements for the GSC datasets. However, the GSC surveys release the underlying ratings on a 0 to 6 scale, which lets us ask the more direct question of how strongly cosine similarity between a participant’s free-text opinion and a candidate statement tracks the rating they actually gave it, which can give us some indication of whether distances in the DPT-tuned and per-topic projected embedding track latent utility as we would hope.

For each of the three GSC surveys, we recover from the raw CSVs every (u, s, r) triple in which participant u rated statement s with Likert score $r \in [0, 6]$. We assemble an anchor pool per participant from all of their free-text responses, which include the open-ended opinion text in the generation and chatbot surveys together with the per-rating justification text that all three surveys collect. The validation cohort writes only justifications. When correlating with the rating of statement s , we exclude the user’s justification text written about s from the anchor pool. The remaining texts are mean-pooled to give a single anchor embedding per rating row. We then compute Spearman rank correlation between cosine similarity and Likert rating, pooled across all rating rows in a survey, under three geometries. These are the base sentence-T5-XL encoder, the DPT-tuned encoder of §5, and the rank-20 per-topic projected embedding $L^\top \psi$ of §7, with L fit from the ideal-point scorer on the same survey’s votes.

Table 12 shows that both adapted geometries lift the pooled correlation substantially over the base encoder on every survey. On the two abortion surveys, which are the most explicitly preference-laden topic in GSC, the per-topic projected embedding more than doubles the chatbot-survey correlation gap and pushes pooled correlation past 0.7 on the validation cohort. The chatbot survey gives the smallest absolute correlations across the board, and it is the one survey on which the global tune beats the per-topic projection.

Table 12: Pooled Spearman rank correlation between cosine similarity (anchor $\rightarrow$ statement) and the participant’s Likert rating, across the three GSC surveys. Higher is better. Best per row in **bold**.

Survey	n ratings	Base	Tuned	Proj
GSC Abortion (gen)	500	0.436	0.614	0.626
GSC Abortion (val)	1000	0.415	0.544	0.721
GSC Chatbot	1100	0.254	0.385	0.331

If we accept that Likert ratings give us a noisy view of the participants’ latent utility, then this experiment provides some evidence that the distances in the DPT-tuned embedding and the per-topic projected embedding are more closely related to utility than distances in the base embedding model.

E.10 Stacking the Per-Topic Probe on the Tuned Encoder

The per-topic metric scorer of §7 is fit on the base sentence-T5-XL encoder. A natural follow-up question is whether the rank-20 projection would perform better if trained on the encoder that has been globally tuned by DPT. We re-run the same val-selected metric probe pipeline on the DPT-tuned encoder, holding all other choices fixed, including the rank, the validation grid, the three-way participant split, and the five seeds.

The natural-test macro is essentially unchanged, moving from 77.6 (base ST5-XL) to 78.0 (DPT-tuned). The hard evaluation triplet performance lifts from 81.1 (base ST5-XL) to 87.4 (DPT-tuned), a 6.3 percentage-point gain, with non-negative gains on every dataset and seven datasets gaining over five points.

Table 13: Per-dataset macro-mean test and 875-triplet hard accuracy of the rank-20 metric probe, fit at val-selected hyperparameters on either the base encoder or the DPT-tuned encoder. Mean and standard deviation over five seeds. Best per row in **bold**.

Dataset	Test (base)	Test (tuned)	Hard (base)	Hard (tuned)
GSC Abortion (gen)	88.3 $\pm$ 0.8	87.5 $\pm$ 0.3	78.0 $\pm$ 0.0	79.2 $\pm$ 0.4
GSC Abortion (val)	85.5 $\pm$ 2.1	86.3 $\pm$ 0.5	82.8 $\pm$ 2.7	86.8 $\pm$ 0.4
GSC Chatbot	76.7 $\pm$ 1.5	76.0 $\pm$ 1.2	67.5 $\pm$ 1.3	77.4 $\pm$ 0.6
Polis Seattle	74.6 $\pm$ 4.0	76.2 $\pm$ 0.8	98.5 $\pm$ 3.4	100.0 $\pm$ 0.0
Polis Bowling Green	72.2 $\pm$ 0.3	73.9 $\pm$ 0.8	84.2 $\pm$ 1.3	93.8 $\pm$ 2.2
Polis Brexit	83.4 $\pm$ 1.7	91.5 $\pm$ 1.0	79.2 $\pm$ 1.5	88.8 $\pm$ 1.9
Polis Canadian	75.7 $\pm$ 1.1	75.5 $\pm$ 1.0	77.4 $\pm$ 1.5	85.2 $\pm$ 1.8
Polis UBI	78.4 $\pm$ 0.7	72.1 $\pm$ 0.3	86.7 $\pm$ 0.0	86.7 $\pm$ 0.0
Remesh Campus	76.9 $\pm$ 0.2	76.5 $\pm$ 0.3	85.4 $\pm$ 1.1	92.2 $\pm$ 0.4
Remesh Foreign	67.0 $\pm$ 0.1	66.5 $\pm$ 0.2	80.8 $\pm$ 1.8	88.0 $\pm$ 0.7
Remesh RTA	74.9 $\pm$ 0.5	76.4 $\pm$ 0.1	71.8 $\pm$ 0.8	83.6 $\pm$ 0.5
Macro mean	77.6	78.0	81.1	87.4

This asymmetry between natural and hard data can be explained using our formal framework from §4.1. The ideal-point scorer projects into a learned rank-20 subspace and never sees the orthogonal complement, so the cosine decomposition of (2) does not literally apply to the probe’s score, but the same mechanism reappears inside the projection. Our hypothesis is that val-selection on natural data tends to pull a few directions into the column space of L that are not strictly preference-aligned, because on natural data those directions correlate positively with preference and so improve validation accuracy. They are nuisance directions that look like signal when semantic and preferential similarity are correlated. When the resulting probe is then evaluated on hard triplets, those nuisance directions contribute against preference, and the projected embedding pays for them with degraded hard-triplet accuracy.

By suppressing the out-of-subspace nuisance globally, the DPT-tuning leaves val-selection with cleaner candidate directions to choose among, and the resulting tuned L contains less of the regime-dependent contamination. We see this directly in the geometry of the learned subspaces. Letting Q_{base} and Q_{tuned} denote orthonormal bases of the two probes’ columns, the singular values of $Q_{\text{base}}^T Q_{\text{tuned}}$ are the cosines of the principal angles between the two rank-20 subspaces. Across the eleven datasets,

the largest cosine averages 0.95 and the median averages 0.80, while the smallest averages 0.17. The two probes share a core preference subspace where most directions are well aligned, and each has a small number of directions that are nearly orthogonal to anything in the other. Those near-orthogonal residues are the encoder-specific component of each fit, and on hard evaluation triplets it is likely the case that the tuned encoder’s residue is benign while the base encoder’s residue carries the nuisance signal that costs accuracy.

A second fingerprint comes from decomposing the score margin as $2\langle L^\top \psi(a), L^\top (\psi(p) - \psi(n)) \rangle + \|L^\top \psi(n)\|^2 - \|L^\top \psi(p)\|^2$, an in-projected-space inner product margin plus a projected item-norm difference, reported per dataset in Table 14. The projected embedding on top of the DPT-tuned model improves hard accuracy on 10 of 11 datasets, and on 8 of those the tuned inner product margin is 1.8 to 5.9 times larger than the base, with the item-norm term staying small in magnitude relative to the inner product margin. The remaining two, Polis Bowling Green and Remesh Foreign, also gain hard accuracy but with only a marginal increase in the mean inner product margin (ratios 1.1 and 1.3 respectively), indicating the improvement there is in where the margin lands rather than its mean. The takeaway is that the same rank-20 budget extracts a stronger preference signal once the base directions have been cleaned by DPT.

Table 14: Per-dataset mean of the in-projected-space inner product margin $2\langle L^\top \psi(a), L^\top (\psi(p) - \psi(n)) \rangle$ and the projected item-norm term $\|L^\top \psi(n)\|^2 - \|L^\top \psi(p)\|^2$ on the dataset’s share of the 875 hard triplets, computed under the rank-20 metric probe fit on either the base or the tuned encoder (single seed). The ratio column is tuned divided by base on the inner product margin.

Dataset	Inner product margin			Item-norm difference	
	Base	Tuned	Ratio	Base	Tuned
GSC Abortion (gen)	+1.30	+2.38	1.8	−0.25	−0.39
GSC Abortion (val)	+1.96	+3.47	1.8	−0.54	−0.61
GSC Chatbot	+0.11	+0.64	5.9	+0.02	+0.10
Polis Seattle	+0.93	+1.94	2.1	+0.08	+0.17
Polis Bowling Green	+1.17	+1.34	1.1	−0.40	−0.09
Polis Brexit	+0.56	+2.17	3.9	−0.02	−0.42
Polis Canadian	+1.16	+5.11	4.4	−0.02	−0.93
Polis UBI	+1.11	+2.00	1.8	−0.18	−0.08
Remesh Campus	+1.79	+3.28	1.8	−0.27	−0.08
Remesh Foreign	+2.17	+2.71	1.3	−0.71	−0.50
Remesh RTA	+0.82	+1.66	2.0	−0.15	+0.07

Two practical readings follow. First, DPT and the per-topic probe are complementary contributions, not redundant. DPT shifts the global encoder geometry to suppress the nuisance subspace, the per-topic projected embedding extracts the dataset-specific preference subspace, and the two compose with no measurable cost on natural data and a consistent gain on hard triplets. Second, val-selection on natural data is not an unalloyed good for downstream evaluation under distribution shift. When the candidate directions a fit can choose from include some that correlate with preference, val-selection can pull them in, and the model will look fine on its own held-out split while degrading on any distribution where the nuisance correlation flips.

F Full Model Comparison

Table 15 is the complete version of the abridged main-results table (Table 4 in §6), covering all 25 embedding models evaluated in this work.

The 25 baselines are: ST5-Base, ST5-Large, ST5-XL [Ni et al., 2022]; all-mpnet-base-v2, all-MiniLM-L6, all-MiniLM-L12, all-distilroberta, NLI-mpnet, Paraphrase-MiniLM [Reimers and Gurevych, 2019]; e5-large-v2 [Wang et al., 2022]; BGE-large-en-v1.5 [Xiao et al., 2024]; GTE-large [Li et al., 2023]; mx-bai-embed-large-v1 [Lee et al., 2024]; Arctic-embed-l [Merrick et al., 2024]; Stella-en-1.5B [Zhang et al., 2024]; Qwen2-1.5B-instruct [Yang et al., 2024]; OpenAI text-embedding-3-small and text-embedding-3-large [OpenAI, 2024]; voyage-3 [Voyage AI, 2024]; voyage-3-large [Voyage AI, 2025]; voyage-4, voyage-4-lite, voyage-4-large [Voyage AI, 2026]; StanceAware-SBERT [Ghafouri et al., 2024]; and BGE-SparseCL (arguana) [Xu et al., 2024].

Table 15: Triplet accuracy (%) across all 11 evaluation datasets, full comparison across 25 models. Best result in each column in **bold**. Abridged version in the body as Table 4.

Model	GSC			Remesh			Polis					Mean
	AbG	AbV	Chat	Camp	Frqn	RTA	Seatl	BG	Brxt	Can	UBI	
ST5-XL + DPT (ours)	81.8	76.4	65.9	67.5	65.6	69.5	69.8	65.1	68.4	62.5	61.9	68.6
ST5-XL	81.1	70.5	69.7	65.8	61.1	69.9	60.7	57.6	56.4	55.5	69.2	65.2
OAI text-emb-3-large	79.6	74.5	56.1	63.9	61.1	71.4	58.1	57.7	51.6	50.0	68.1	62.9
ST5-Large	75.5	65.5	57.2	64.6	59.9	69.2	58.8	58.1	49.9	56.8	72.2	62.5
Voyage-4-large	69.9	69.0	63.7	60.0	57.9	69.0	55.4	58.3	50.3	48.3	63.6	60.5
e5-large-v2	63.9	63.6	51.8	65.1	59.4	70.5	53.5	57.1	56.2	57.5	66.1	60.4
Voyage-4	72.3	66.6	62.1	58.4	56.7	67.0	54.9	57.4	50.8	50.1	64.5	60.1
ST5-Base	68.6	62.5	47.1	63.7	59.0	66.6	58.4	56.9	51.1	56.0	68.8	59.9
NLI-mpnet	56.1	66.9	47.6	65.6	61.0	69.5	58.3	56.9	52.8	55.5	68.7	59.9
MxBAl-embed-large	63.9	62.4	55.1	62.7	59.9	69.2	54.2	56.2	50.5	54.5	70.0	59.9
GTE-large	67.6	65.6	59.5	60.2	56.3	67.7	54.7	55.5	46.2	51.6	69.5	59.5
OAI text-emb-3-small	63.8	65.3	51.9	62.1	58.2	70.8	56.2	57.0	50.3	50.4	66.0	59.3
Voyage-3-large	68.3	61.0	60.6	57.2	57.5	67.6	58.5	57.6	50.7	49.8	63.5	59.3
Voyage-4-lite	70.1	64.5	61.3	58.2	56.8	65.4	53.3	57.0	51.2	49.3	65.5	59.3
BGE-large-en	64.7	59.5	56.5	62.5	57.6	68.0	54.2	55.5	53.2	52.9	66.3	59.2
StanceAware-SBERT	58.5	58.2	36.4	64.8	59.7	68.2	68.1	56.1	52.5	52.3	68.3	58.5
Paraphrase-MiniLM	58.2	67.8	51.7	59.7	58.3	67.5	51.7	56.4	53.3	52.4	64.7	58.3
all-mpnet-base	57.1	55.6	60.4	58.5	57.0	69.2	54.4	56.4	47.1	52.4	68.0	57.8
Arctic-embed-l	56.4	62.4	58.5	57.6	56.2	67.5	53.4	57.6	50.2	47.3	64.2	57.4
all-distilroberta	61.8	55.8	53.4	59.5	57.5	67.6	53.7	55.4	50.5	48.5	65.8	57.2
BGE-SparseCL (arguana)	62.4	61.7	42.7	59.2	57.2	66.1	50.1	56.1	57.5	52.6	63.0	57.2
Voyage-3	63.8	50.4	57.5	58.0	54.9	65.2	53.0	55.5	50.6	45.8	65.2	56.4
all-MiniLM-L12	57.8	57.1	47.6	59.2	57.8	67.2	52.0	55.3	49.9	48.8	64.1	56.1
all-MiniLM-L6	55.4	52.3	48.9	60.2	57.6	67.1	53.6	55.1	48.4	48.1	65.4	55.7
Qwen2-1.5B-instruct	59.6	50.8	49.7	56.5	54.9	58.0	48.5	50.9	49.4	50.5	62.7	53.8
Stella-en-1.5B	43.5	50.2	65.3	55.0	55.1	54.0	53.4	49.1	47.4	57.0	49.7	52.7

F.1 Baseline encoding format

Many baselines were trained as asymmetric retrieval encoders, expecting the query side and the passage side of an input to be formatted differently. Treating these encoders symmetrically (passing both anchor and candidate through plain `encode(text)`) consistently under-represents what they can do. We therefore apply each model’s intended query/passage convention, treating the participant’s anchor text as the query and the candidate statement as the passage. The conventions below are sourced from each model card or registered `config_sentence_transformers.json`, and in all cases cosine similarity is computed between the query-side and passage-side embeddings.

- **Symmetric (no prefixes):** ST5-XL/Large/Base, all-mpnet-base, all-MiniLM-`{L6,L12}`, all-distilroberta, NLI-mpnet, Paraphrase-MiniLM, GTE-large [Li et al., 2023] (the older non-instruct variant), OpenAI text-embedding-3-`{small, large}`, StanceAware-SBERT.
- **e5-large-v2:** "query: " prefix on anchors, "passage: " prefix on candidates.
- **BGE-large-en-v1.5, MxBAl-embed-large-v1:** "Represent this sentence for searching relevant passages: " prefix on anchors, plain on candidates.
- **Snowflake Arctic-embed-l-v2.0:** "query: " prefix on anchors, plain on candidates. The v2 redesign uses E5-style prefixes rather than the v1 BGE prompt.
- **BGE-SparseCL (arguana):** plain encoding on both sides, matching the authors’ own evaluation script. The released checkpoint ships without an `SBERT modules.json`; we verified that the default sentence-transformers wrapper applies mean pooling, matching the paper’s training-time `--pooler_type avg`.
- **Voyage:** the API exposes an `input_type` parameter; we pass "query" for anchors and "document" for candidates.
- **Instruction-tuned (GTE-Qwen2-1.5B-instruct, Stella-en-1.5B-v5):** task-tailored instruction wrapped in each model’s standard "Instruct: <task>\nQuery: " template on anchors, plain text on candidates.

Instruction-tuned prompt selection. The registered prompts in each model’s `config_sentence_transformers.json` target web-search retrieval (e.g., “Given a web search query, retrieve relevant passages that answer the query”), which does not match a sentence-to-sentence preference task. Qwen was trained with diverse task instructions and supports arbitrary task strings. Stella was trained primarily on two registered prompts, `s2p_query` for

sentence-to-passage retrieval and s2s_query for sentence-to-sentence similarity, with the model card recommending those two for general use. We swap in a task-tailored instruction, “*Given an opinion statement on a contested social or political issue, retrieve other statements that share the same stance and underlying values,*” using the template described above.

For Stella, we verified that this choice is benign on natural data: the task-tailored instruction yields a 52.7% macro-mean, the registered s2s_query prompt yields 52.8% (a 0.1 pp difference), and the s2p_query prompt yields 49.4% (worse, as expected for the web-search framing). For Qwen, the task-tailored instruction and the registered web-search instruction give identical 53.8% macro-mean. The Table 15 rows for these two models report the task-tailored configuration.

G Prompts

Three LLM prompts define the training-data pipeline: GPT-4o-mini filters the issue pool, Claude Sonnet generates 5 diverse opinions per surviving issue, and GPT-4o rewrites a sampled anchor into a hard triplet. The same rewrite prompt is used to produce the held-out hard eval triplets from real participant anchors (§4.3). Placeholders in braces ({issue_text}, {anchor}) are substituted at call time. For reproducibility, exact model identifiers and sampling settings are:

- **Issue filter:** gpt-4o-mini, temperature 0, max tokens 5.
- **Opinion generation:** claude-sonnet-4-20250514, temperature 1.0, max tokens 1024.
- **Hard-triplet rewrite:** gpt-4o, temperature 0.7, max tokens 400.

Issue filter (GPT-4o-mini, one call per candidate issue).

You are deciding whether a debate topic is a political, policy, or social issue suitable for studying public opinion and preference diversity.

ACCEPT if:

- It is a political, policy, economic, or social issue
- People’s positions reflect their values, ideology, or worldview (not just personal taste)
- It is the kind of topic debated in legislatures, newspapers, public surveys, or town halls
- There is a clear spectrum of positions (e.g., pro-regulation vs. free-market)

REJECT if:

- It is purely factual or scientific with a clear answer
- It is abstract philosophy with no policy relevance
- It is a niche technical question that only specialists would engage with
- It is about personal lifestyle preferences rather than public affairs
- It has near-universal consensus
- It is nonsensical or too vague

Topic: "{issue_text}"

Respond with exactly one word: ACCEPT or REJECT

Opinion generation (Claude Sonnet, one call per issue).

Topic: {issue_text}

Write 5 standalone opinions on this topic from 5 different people. These should read like things people wrote in a forum - NOT like they are answering a question.

The 5 opinions should be clearly distinct and evenly spaced across the full opinion spectrum:

1. Strongly supportive - fully committed to this position
2. Moderately supportive - in favor but with reservations
3. Genuinely ambivalent - sees valid points on both sides

4. Moderately opposed - against but acknowledges some merit
5. Strongly opposed - firmly against

Important:

- Do not start with Yes, No, I agree, I disagree or any response-like framing
- Each opinion should stand alone as a statement of belief
- Make the difference between adjacent positions (e.g., 1 vs 2, 4 vs 5) clear and meaningful
- Keep them short and natural - average ~25 words, some shorter, some longer

```
{"opinions": ["op1", "op2", "op3", "op4", "op5"]}
```

Hard-triplet rewrite (GPT-4o, one call per sampled anchor).

You are given a person's opinion statement on a political or social topic.

Original statement: "{anchor}"

Generate two rewritten versions that test whether a model understands values vs. word overlap:

1. SEMANTIC DISTRACTOR (tricks models that rely on word overlap):
 - Keep nearly IDENTICAL wording and sentence structure as the original
 - Only flip the conclusion/stance to be the opposite
 - Goal: MAXIMIZE word overlap while having opposite meaning
 - Example: "Renewable energy must be prioritized above all else"
 - > "Renewable energy is worth considering, but should NOT be prioritized above all else"
2. PREFERENCE MATCH (requires understanding values, not words):
 - Express the SAME stance and values as the original
 - But use completely DIFFERENT vocabulary, framing, and sentence structure
 - Goal: MINIMIZE word overlap while preserving the underlying position
 - Example: "Renewable energy must be prioritized above all else"
 - > "Fossil fuel dependence is our greatest threat - government must act decisively on clean power"

Respond with valid JSON only:

```
{"semantic_distractor": "...", "preference_match": "..."}

```

Normal-correlation rewrite (GPT-4o, one call per sampled anchor).

You are given a person's opinion statement on a political or social topic.

Original statement: "{anchor}"

Generate two rewritten versions that exhibit the natural correlation between stance and wording, where same stance tends to share words and opposite stance tends to use different words:

1. PREFERENCE MATCH (high word overlap with the anchor, same stance):
 - Reuse most of the anchor's vocabulary and content words
 - Preserve the original stance and values
 - BUT it must be a genuine paraphrase, NOT a verbatim or near-verbatim copy of the anchor. Change at least 3-5 words to synonyms, rearrange clauses, or restructure the sentence. Otherwise the rewrite is invalid.
 - Goal: MAXIMIZE word overlap while still meaningfully rewording
 - Example: "Renewable energy must be prioritized above all else"
 - > "We must put renewable energy ahead of every other concern"

2. SEMANTIC DISTRACTOR (low word overlap with the anchor, opposite stance):
- Use completely DIFFERENT vocabulary, framing, and sentence structure
 - Express the OPPOSITE stance on the same underlying issue
 - Goal: MINIMIZE word overlap while having opposite meaning
 - Example: "Renewable energy must be prioritized above all else"
-> "Cheap, reliable fossil fuels are what people actually need. Chasing wind and solar comes at too great a cost."

Respond with valid JSON only:

```
{"preference_match": "...", "semantic_distractor": "..."}

```